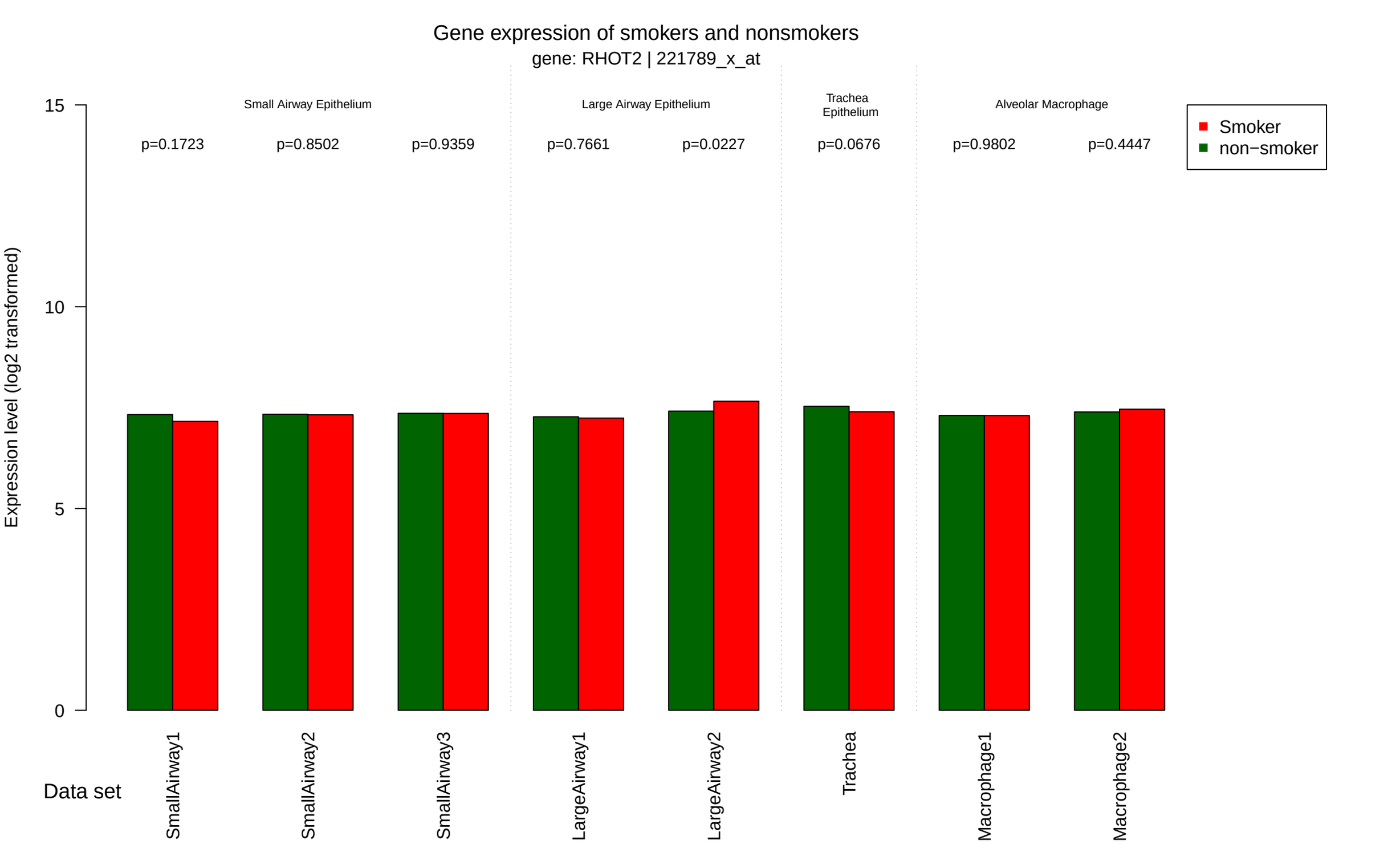
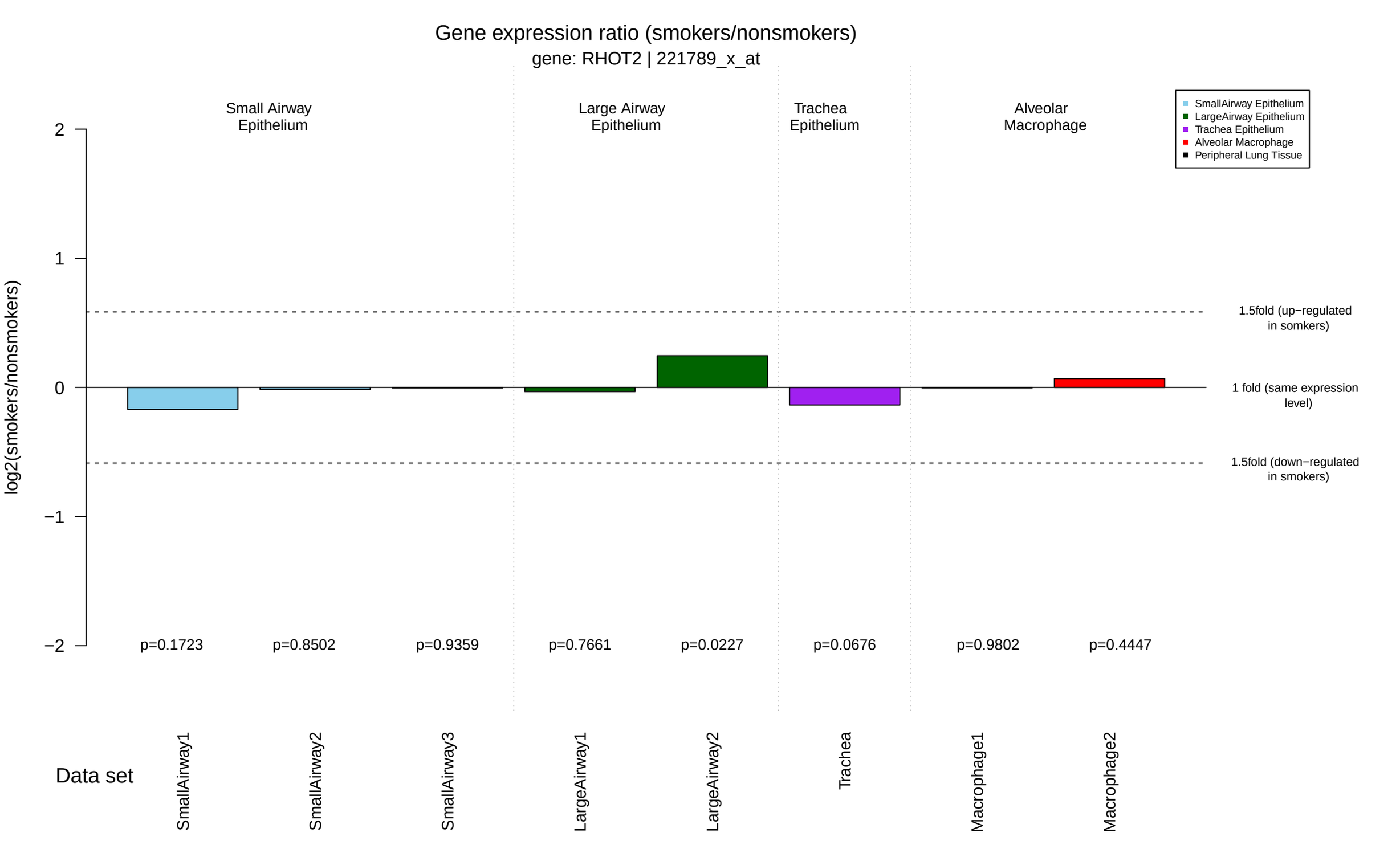
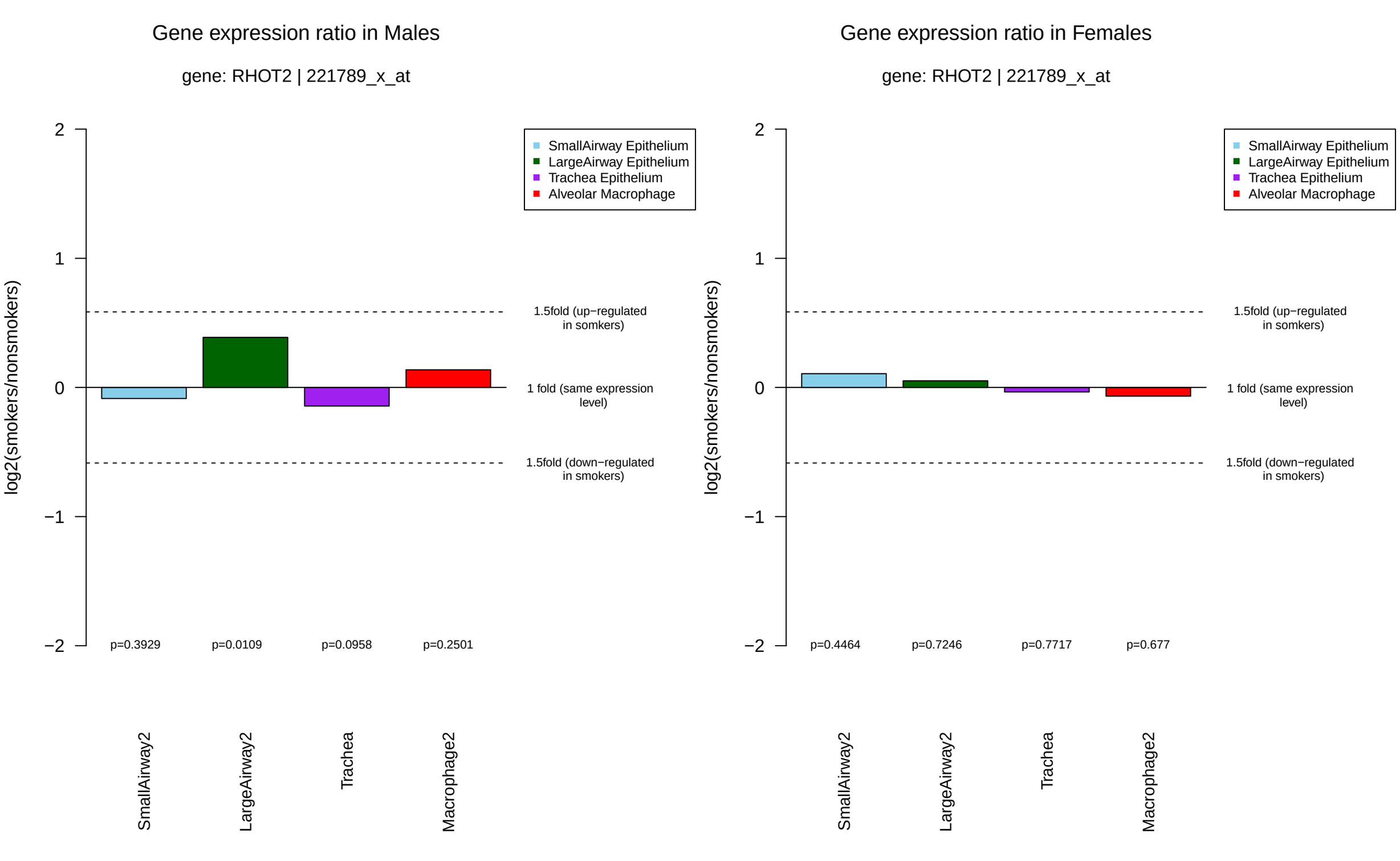
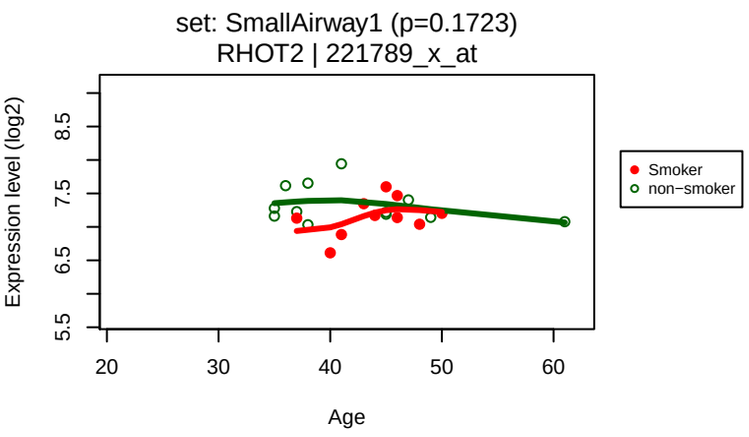
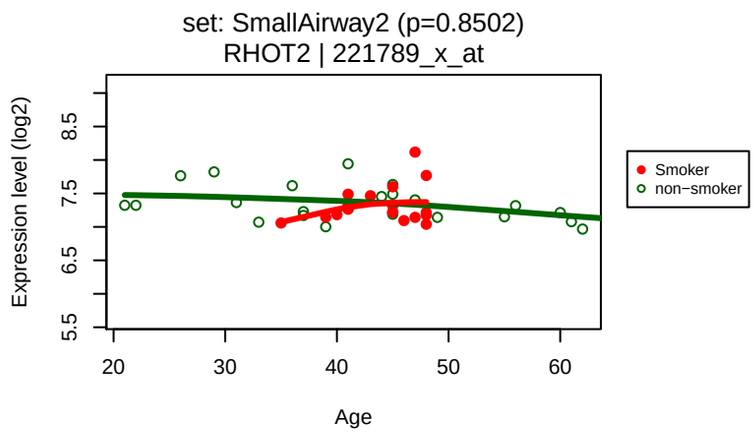
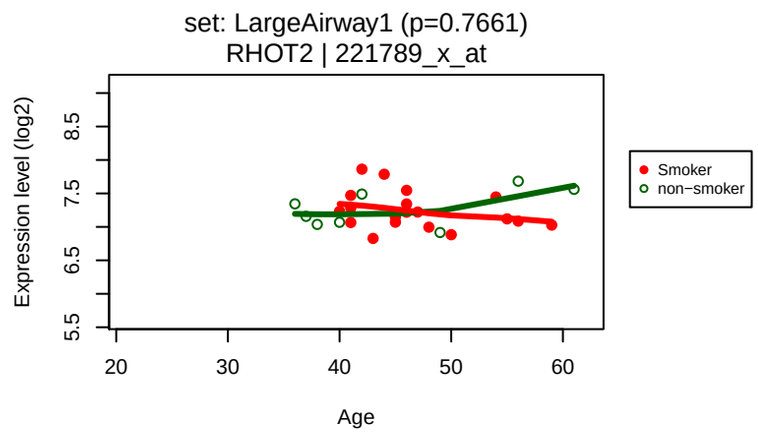

Gene expression of smokers and nonsmokers
gene: RHOT2 | 221789_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1723
p=0.8502
p=0.9359
p=0.7661
p=0.0227
p=0.0676
p=0.9802
p=0.4447
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

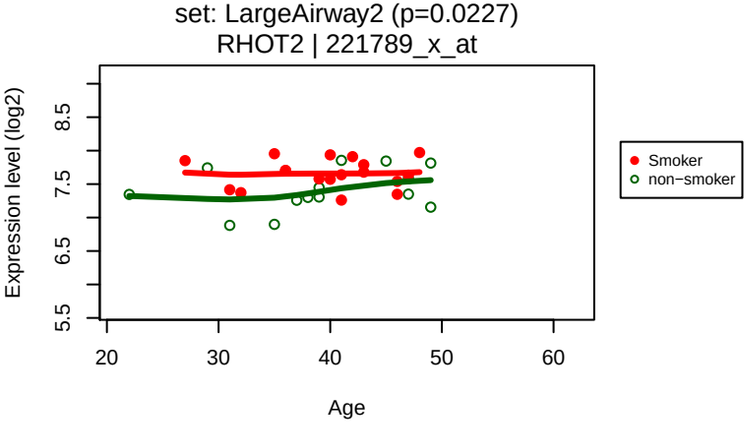

Gene expression ratio (smokers/nonsmokers)
gene: RHOT2 | 221789_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1723
p=0.8502
p=0.9359
p=0.7661
p=0.0227
p=0.0676
p=0.9802
p=0.4447
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

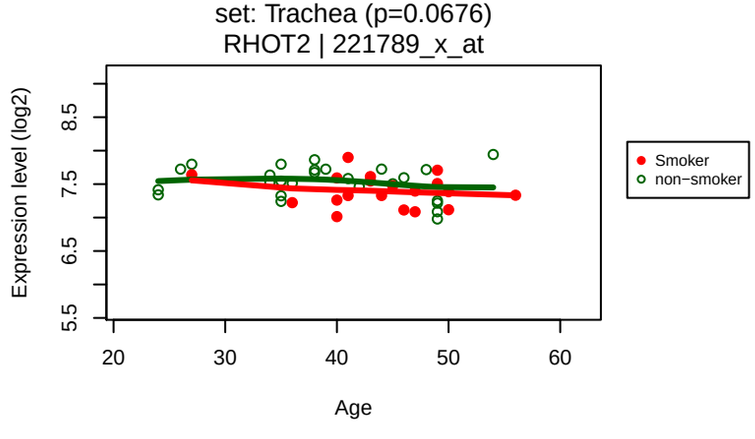

Gene expression ratio in Males
gene: RHOT2 | 221789_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3929
p=0.0109
p=0.0958
p=0.2501
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHOT2 | 221789_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4464
p=0.7246
p=0.7717
p=0.677
SmallAirway2
LargeAirway2
Trachea
Macrophage2

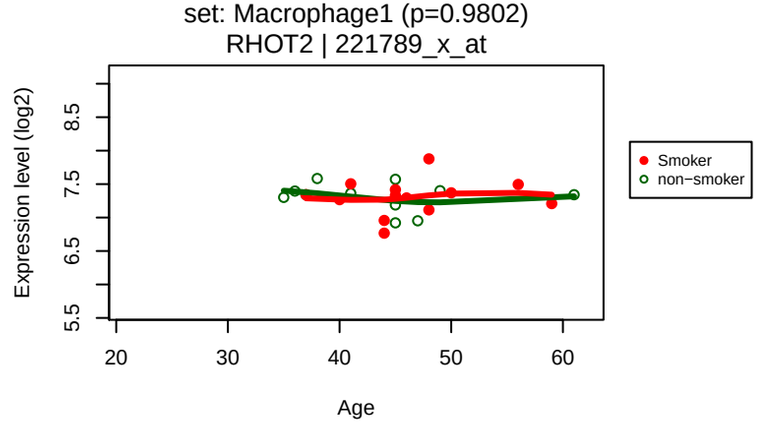

set: SmallAirway1 (p=0.1723)
RHOT2 | 221789_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

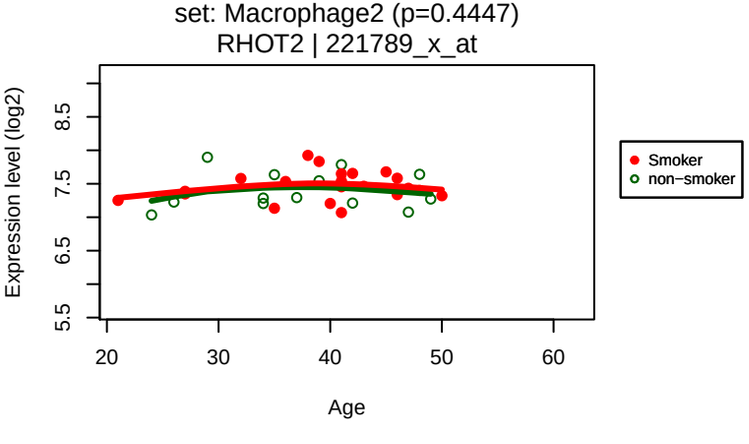

set: SmallAirway2 (p=0.8502)
RHOT2 | 221789_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

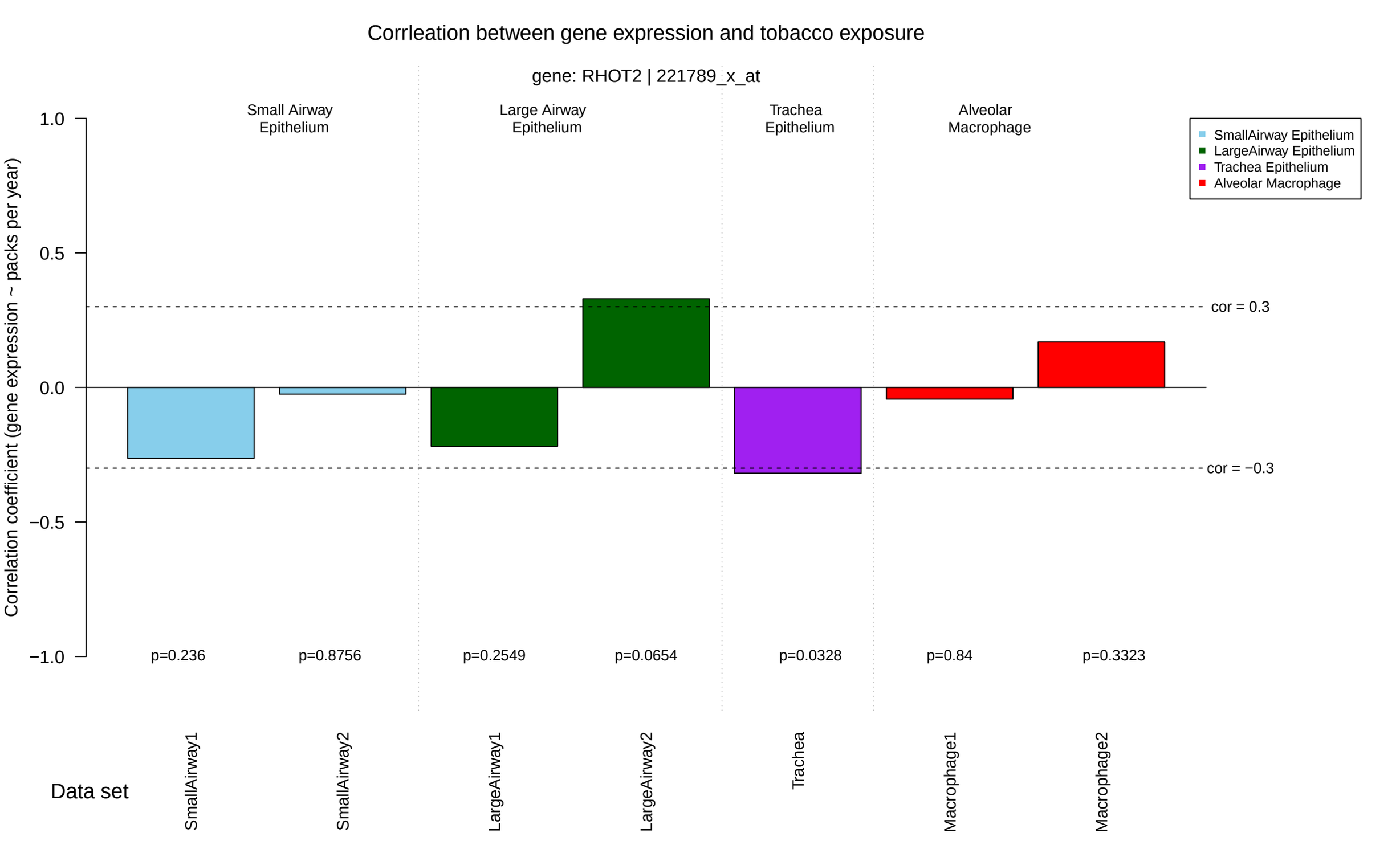

set: LargeAirway1 (p=0.7661)
RHOT2 | 221789_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0227)
RHOT2 | 221789_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0676)
RHOT2 | 221789_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9802)
RHOT2 | 221789_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4447)
RHOT2 | 221789_x_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHOT2 | 221789_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.236
p=0.8756
p=0.2549
p=0.0654
p=0.0328
p=0.84
p=0.3323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2